Task-1

Data Cleaning & Preprocessing

1. Objective

To transform the raw Netflix dataset into a clean, structured format by addressing null values, duplicates, and inconsistent data types using Python (Pandas).

2. Tools Used

- **Language:** Python
- **Library:** Pandas
- **Environment:** Google Colab / Jupyter Notebook

3. Implementation of the Core Concepts

Identify and Handle Missing Values

Description: Used .isnull().sum() to locate missing data. Critical columns like director and cast had significant nulls, which were filled with "Unknown" to maintain data volume.

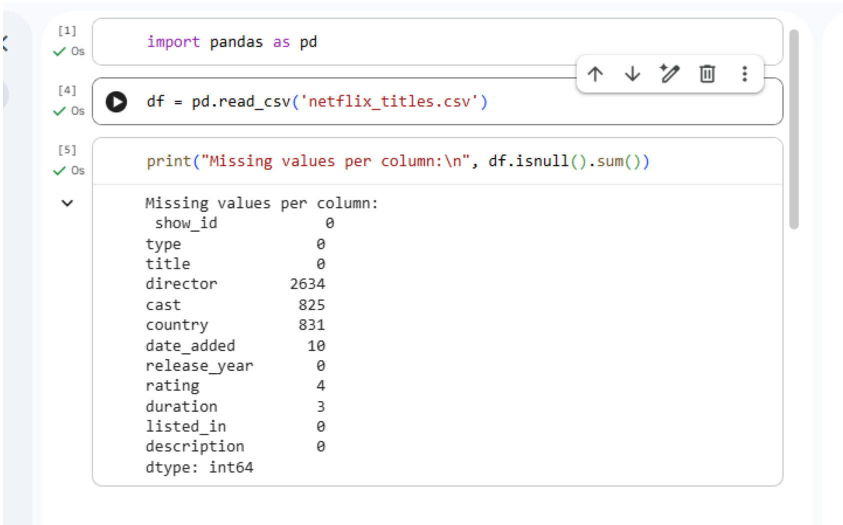
Coding:

```
import pandas as pd

# Load dataset
df = pd.read_csv('netflix_titles.csv')
# Identify missing values
print("Missing values per column:\n", df.isnull().sum())
```

Fill missing categorical values
df['director'] = df['director'].fillna('Unknown')
df['cast'] = df['cast'].fillna('Unknown')
df['country'] = df['country'].fillna('Not Specified')
# Drop rows where critical info like 'date_added' or 'rating' is missing
df.dropna(subset=['date_added', 'rating', 'duration'], inplace=True)

Screenshot:



```
[1] import pandas as pd

[4] df = pd.read_csv('netflix_titles.csv')

[5] print("Missing values per column:\n", df.isnull().sum())

Missing values per column:
 show_id           0
type              0
title             0
director       2634
cast            825
country         831
date_added       10
release_year      0
rating            4
duration          3
listed_in         0
description       0
dtype: int64
```

♾ Remove Duplicate Rows

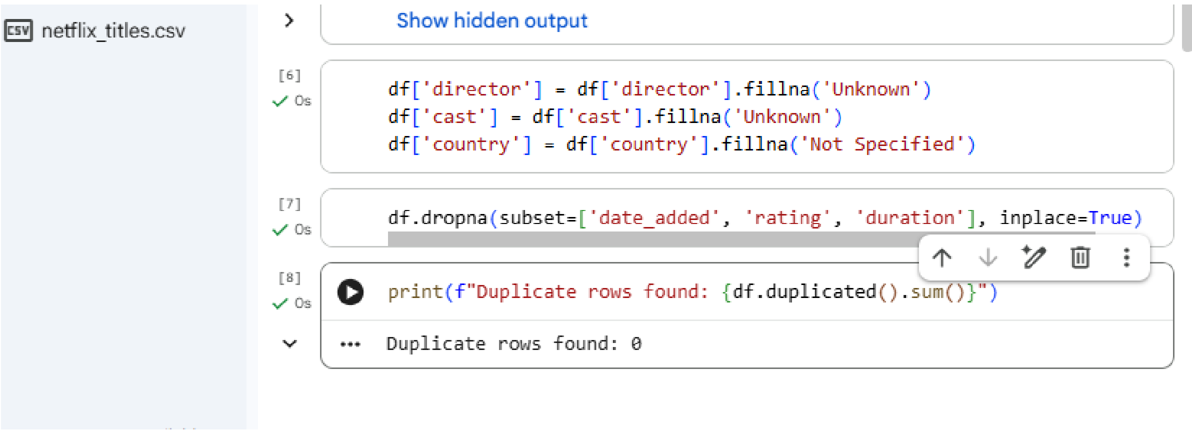
Description: Duplicates can lead to incorrect statistics. I used .drop_duplicates() to ensure every row represents a unique title.

Coding:

# Check for duplicates
print(f"Duplicate rows found: {df.duplicated().sum()}")

# Remove duplicates
df.drop_duplicates(inplace=True)

Screenshot:



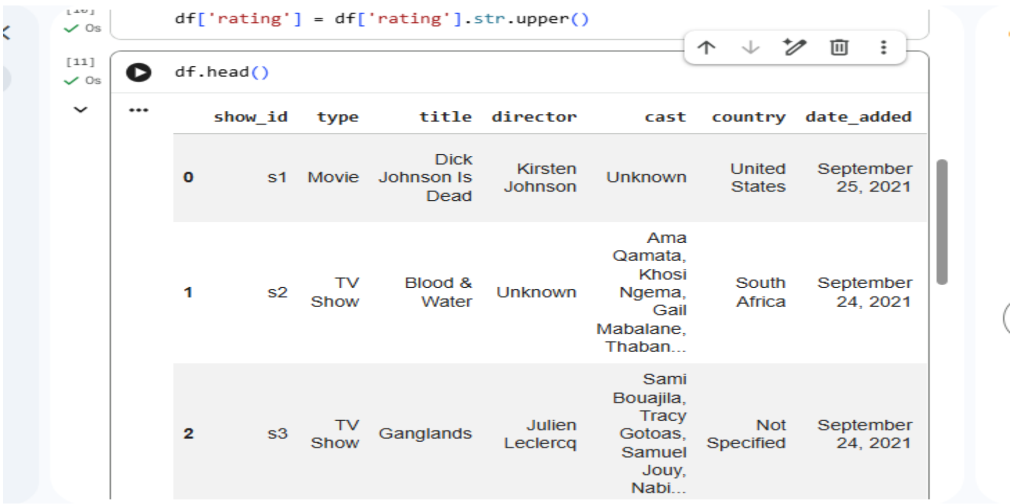
Standardize Text Values

Description: Standardized the type (Movie/TV Show) and rating columns by removing leading/trailing spaces and ensuring consistent casing.

Coding:

```
# Strip extra spaces from the 'type' and 'title' columns
df['type'] = df['type'].str.strip()
df['title'] = df['title'].str.strip()
# Standardize 'rating' to uppercase
df['rating'] = df['rating'].str.upper()
```

Screenshot:



Convert Date Formats

Description: The date_added column was a string. I converted it to a standard datetime format (YYYY-MM-DD) to allow for time-series analysis.

Coding:

```
# Remove leading/trailing spaces from the string date
df['date_added'] = df['date_added'].str.strip()
# Convert to datetime format
df['date_added'] = pd.to_datetime(df['date_added'])
```

Screenshot:

```
df['date_added'] = df['date_added'].str.strip()
df['date_added'] = pd.to_datetime(df['date_added'])
```

```
df[['title', 'date_added']].head()
```

	title	date_added
0	Dick Johnson Is Dead	2021-09-25
1	Blood & Water	2021-09-24
2	Ganglands	2021-09-24
3	Jailbirds New Orleans	2021-09-24
4	Kota Factory	2021-09-24

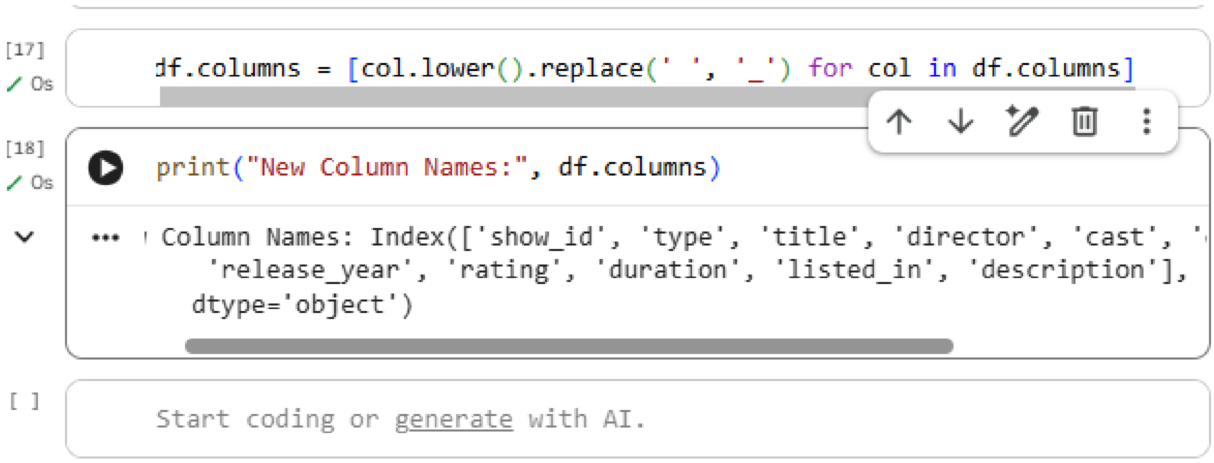
Rename Column Headers

Description: Renamed headers to follow the snake_case convention (lowercase with underscores) for better compatibility with SQL and coding standards.

Coding:

```
# Rename columns to be clean and uniform
df.columns = [col.lower().replace(' ', '_') for col in df.columns]
# Example: If you want to rename a specific column
df.rename(columns={'listed_in': 'category'}, inplace=True)
print("New Column Names:", df.columns)
```

Screenshot:



Check and Fix Data Types

Description: Verified that numeric columns like release_year are integers and the newly converted date columns are recognized as datetime objects.

Coding:

```
# Check data types
print("Data types before fix:\n", df.dtypes)
# Ensure release_year is an integer
df['release_year'] = df['release_year'].astype(int)
# Final check
print("Data types after fix:\n", df.dtypes)
```

Save the cleaned data

df.to_csv('cleaned_netflix_data.csv', index=False)

print("\n--- Project Task 1 Completed Successfully! ---") print(df.head())

Screenshots:

```
[19]
print("Data types before fix:\n", df.dtypes)

Data types before fix:
 show_id                 object
type                    object
title                   object
director                object
cast                    object
country                 object
date_added      datetime64[ns]
release_year             int64
rating                  object
duration                object
listed_in               object
description             object
dtype: object
```

```
[20]
df['release_year'] = df['release_year'].astype(int)
print("Data types after fix:\n", df.dtypes)

Data types after fix:
 show_id                 object
type                    object
title                   object
director                object
cast                    object
country                 object
date_added      datetime64[ns]
release_year             int64
rating                  object
duration                object
listed_in               object
description             object
dtype: object
```

[22]

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 8790 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       8790 non-null   object
 1   type          8790 non-null   object
 2   title         8790 non-null   object
 3   director      8790 non-null   object
 4   cast          8790 non-null   object
 5   country       8790 non-null   object
 6   date_added    8790 non-null   datetime64[ns]
 7   release_year  8790 non-null   int64
 8   rating        8790 non-null   object
 9   duration      8790 non-null   object
 10  listed_in     8790 non-null   object
 11  description   8790 non-null   object
dtypes: datetime64[ns](1), int64(1), object(10)
memory usage: 892.7+ KB
```

[23]

```
print("\n--- Project Task 1 Completed Successfully! --- ")
print(df.head())
```

```

--- Project Task 1 Completed Successfully! ---
  show_id     type                 title         director  \
0      s1    Movie  Dick Johnson Is Dead  Kirsten Johnson
1      s2  TV Show         Blood & Water          Unknown
2      s3  TV Show             Ganglands  Julien Leclercq
3      s4  TV Show  Jailbirds New Orleans          Unknown
4      s5  TV Show          Kota Factory          Unknown

                                                cast        country  \
0                                            Unknown  United States
1  Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...   South Africa
2  Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...  Not Specified
3                                            Unknown  Not Specified
4  Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...          India

  date_added  release_year rating   duration  \
0 2021-09-25          2020  PG-13     90 min
1 2021-09-24          2021  TV-MA  2 Seasons
2 2021-09-24          2021  TV-MA   1 Season
3 2021-09-24          2021  TV-MA   1 Season
4 2021-09-24          2021  TV-MA  2 Seasons
```